

# Shubham Mahesh Mohole

+1 (716) 398 9180 — shumoh2000@gmail.com — Portfolio — LinkedIn — GitHub

**Software Engineer** specializing in high-throughput Spring Boot microservices and distributed systems. Delivered production systems handling **5M+ daily transactions** at Infosys with measurable latency and throughput wins. Completed M.S. in Computer Science at University at Buffalo (Dec 2025); seeking full-time SWE/SDE.

## EDUCATION

### University at Buffalo, SUNY

M.S. in Computer Science

GPA: 3.7/4

Buffalo, NY, USA

Aug 2024 – Dec 2025

**Relevant Courses:** Algorithms, Machine Learning, Computer Vision, Database Systems, Deep Learning, Operating Systems

### University of Pune

Bachelor of Engineering

GPA: 8.7/10

Pune, MH, India

Aug 2018 – May 2022

## SKILLS

- **Languages:** Python, Java, C++, SQL, JavaScript
- **Backend & Systems:** Microservices, REST APIs, Distributed Systems, Event-Driven Architecture, gRPC, Rate Limiting, Circuit Breakers, Consistent Hashing, Idempotency
- **Core CS:** Data Structures & Algorithms, Concurrency, Multithreading, OOP, Thread Safety, JVM Internals, Memory Management, GC Tuning
- **Databases:** MySQL, Redis (caching, Lua scripting, pub/sub), MongoDB, NoSQL, Query Optimization, Indexing
- **Cloud & DevOps:** AWS, Docker, Kubernetes, CI/CD, Jenkins, Containerization, Zero-downtime Deployments

## PROFESSIONAL EXPERIENCE

### University at Buffalo, SUNY — Robotics & AI Lab (Advisor: Prof. Nalini Ratha)

Research Assistant

Buffalo, NY, USA

Aug 2025 – Present

- Building modular Python experimentation pipelines to benchmark perception model robustness across simulated and real sensor environments; targeting submission to a computer vision venue.
- Automated data preprocessing workflows, cutting experiment turnaround time by ~40% and improving result reproducibility across team members.

### Infosys Ltd.

Software Developer — Systems Engineer

Pune, India

Nov 2022 – May 2024

- Owned design and development of Java 17 / Spring Boot microservices supporting **5M+ daily transactions** for a financial-services client, ensuring correctness and fault tolerance under peak load.
- Cut onboarding latency by **25%** by redesigning database access patterns, connection pooling, and thread utilization; reduced P99 latency to **40ms**.
- Boosted throughput by **90%** via MySQL query optimization, composite indexing strategies, and Redis-based distributed caching (sliding-window rate limiting + TTL eviction).
- Engineered resilient REST integrations with external financial APIs using exponential-backoff retries, circuit breakers (Resilience4j), and structured exception handling, achieving 99.9% uptime SLA.
- Containerized and deployed services on Kubernetes with rolling-update CI/CD pipelines (Jenkins), enabling zero-downtime releases and rapid rollback capability.
- Eliminated memory pressure hot spots through JVM heap profiling and GC tuning (G1GC), stabilizing latency under sustained load.
- Led load testing and chaos-engineering exercises to validate system behavior at 2× peak traffic; findings drove three architectural improvements before production launch.

### PMS Robotics Lab

Software Engineer Intern

Pune, India

May 2022 – Nov 2022

- Developed and validated autonomous navigation algorithms in Python and C++ via ROS-based simulation, improving path-tracking accuracy by 15% through systematic parameter tuning and multi-scenario regression testing.

## PROJECTS

### Intelligent Distributed Task Queue Engine — [GitHub](#)

April 2026

- Engineered a distributed async job processing system with at-least-once delivery, idempotent submissions, and retry + DLQ handling using Redis and Lua scripting.
- Implemented ML-based scheduling with GradientBoostingRegressor and safe FIFO fallback, improving average job completion latency by **5.44%**.
- Achieved **100% success rate**, **11ms P95 submission** and **~2.0s P95 completion** latency under k6 load tests.
- Designed priority queues using Redis Sorted Sets + Lists and built observability stack with Prometheus and Grafana.

### Distributed Rate Limiter — [GitHub](#)

Jan 2026

- Built a distributed rate limiter with atomic Redis Lua enforcement supporting token bucket, fixed window, sliding window log, and sliding window counter algorithms.
- Analyzed and fixed performance regression due to profiling overhead and connection pool contention in high-concurrency scenarios.
- Achieved **2.9K req/s** with **0% error rate** and **p95 latency ~372ms** under stable k6 benchmarks in Docker.
- Optimized hot-path execution by caching Lua, tuning connection pools, and eliminating unnecessary serialization overhead.

### AI Health Screening Platform — [Live](#)

April 2026

- Developed a full-stack AI health screening platform with symptom analysis and image-based diagnosis using ML models.
- Implemented JWT authentication, REST APIs, and PostgreSQL persistence for user assessments and history tracking.
- Integrated Random Forest and ResNet18 (PyTorch) models with preprocessing pipelines and confidence scoring.
- Deployed with frontend on Vercel and backend on Render using Dockerized production environments.